

Refinery Maintenance Patterns: A Multi-Regional Analysis of Petroleum Markets

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Abstract—This paper presents a novel trading strategy leveraging refinery maintenance patterns across different Petroleum Administration for Defense Districts (PADD) regions to generate profitable trading opportunities in the transportation sector. Through iterative development and backtesting, we demonstrate how identifying maintenance-driven market inefficiencies, combined with sophisticated position sizing and risk management, can yield significant returns. The final strategy achieved an 84.62% win rate and generated \$340,522 in profit from a \$1 million initial investment over a two-year period.

Index Terms—petroleum markets, trading strategy, refinery maintenance, risk management, position sizing, regional analysis

I. INTRODUCTION

The petroleum refining industry exhibits predictable maintenance patterns that can create regional price disparities in gasoline markets. This research explores how these patterns can be exploited through a systematic trading approach focused on transportation company stocks. We analyze data across five PADD regions and develop a trading strategy that evolves through multiple iterations, incorporating increasingly sophisticated risk management and position sizing techniques.

II. DATA COLLECTION AND METHODOLOGY

A. Data Sources

The study utilizes three primary data sets:

- Refinery utilization rates from the Energy Information Administration (EIA)
- Regional gasoline price data
- Stock price data for major transportation companies

B. Regional Coverage

The analysis covers all five PADD regions:

- PADD 1: East Coast
- PADD 2: Midwest
- PADD 3: Gulf Coast
- PADD 4: Rocky Mountain
- PADD 5: West Coast

C. Time Period

The study spans from January 2022 to December 2023, providing a comprehensive dataset across different seasonal patterns and market conditions.

III. PATTERN RECOGNITION AND ANALYSIS

A. Maintenance Pattern Identification

Maintenance periods were identified using the following criteria:

- 1) Utilization rates below regional mean minus one standard deviation
- 2) Rolling average analysis with 4-week windows
- 3) Seasonal pattern consideration

B. Price Impact Analysis

Key findings in price impact analysis include:

TABLE I
REGIONAL PRICE CHANGES DURING MAINTENANCE

Region	Average Change (%)
PADD 1	1.83
PADD 2	2.37
PADD 3	4.64
PADD 4	6.57
PADD 5	3.50

1) Average Price Changes During Maintenance:

TABLE II
DAYS TO MAXIMUM PRICE IMPACT

Region	Days to Max
PADD 1	1.47
PADD 2	0.89
PADD 3	1.52
PADD 4	2.47
PADD 5	2.22

2) Price Impact Timing:

IV. STRATEGY DEVELOPMENT

A. Initial Strategy

The baseline strategy implemented simple maintenance period identification with basic position sizing:

- Total Trades: 25
- Win Rate: 60.00%
- Average Return: 0.69%
- Total PnL: \$54,108.41

B. Improved Strategy

The second iteration introduced more sophisticated position sizing and regional specialization:

- Total Trades: 12
- Win Rate: 75.00%
- Average Return: 2.75%
- Total PnL: \$140,472.85
- Sharpe Ratio: 0.63
- Max Drawdown: \$-7,908.70

C. Final Strategy

The final version incorporated dynamic position sizing and enhanced risk management:

- Total Trades: 13
- Win Rate: 84.62%
- Average Return: 3.49%
- Total PnL: \$340,522.53
- Sharpe Ratio: 0.63
- Max Drawdown: \$-14,884.04

D. Machine Learning Enhancement

An additional iteration incorporating machine learning techniques showed:

- Total Trades: 6
- Win Rate: 83.33%
- Average Return: 2.74%
- Total PnL: \$101,208.80
- Sharpe Ratio: 1.20

V. RISK MANAGEMENT FRAMEWORK

A. Position Sizing Rules

The final strategy implemented dynamic position sizing based on:

- 1) Confidence metrics
- 2) Regional performance history
- 3) Seasonal factors
- 4) Rolling profit/loss

B. Risk Controls

1) Maximum Position Sizes:

- PADD 4: 40%
- PADD 3: 30%
- PADD 5: 25%
- PADD 2: 20%

2) Stop-Loss Mechanisms:

- Portfolio-level stop at 5% drawdown
- Dynamic trailing stops based on profit levels
- Region-specific maximum loss limits

VI. RESULTS AND ANALYSIS

A. Performance Metrics

Final strategy key performance indicators:

- Portfolio Growth: 34.05%
- Win Rate: 84.62%
- Risk-Adjusted Return (Sharpe): 0.63
- Maximum Drawdown: -1.49%

B. Regional Performance

Performance varied significantly by region:

- 1) PADD 4 showed highest return potential but greater volatility
- 2) PADD 3 demonstrated most consistent performance
- 3) PADD 1 was eventually excluded due to poor risk-adjusted returns

C. Seasonal Patterns

The strategy showed strongest performance during:

- 1) Spring (April-May): Highest return period
- 2) Early Winter (December): Strong consistent returns
- 3) Mid-Summer (July): Most challenging period

VII. CONCLUSION

A. Key Findings

- 1) Maintenance patterns provide actionable trading signals
- 2) Regional specialization improves performance
- 3) Dynamic position sizing enhances returns
- 4) Sophisticated risk management is crucial for consistency

B. Future Research Directions

Future work could focus on:

- 1) Integration of additional data sources
- 2) Enhancement of machine learning components
- 3) Extension to other energy-dependent sectors
- 4) Development of real-time signal generation